

Design of an Instrumented Robot Data-Collection Cell for Physical AI

ME 423-X2

Individual Concept and Subsystem Analysis

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"I pledge my honor that I have abided by the Stevens Honor System."

Subsystem

The subsystem being analyzed is the mass distribution of the handheld unit, which is affected by the location of the handle, the angle of components attached to the system, and the infill percentage of the 3D-printed components. The mass distribution will be used to determine the sustained wrist and forearm load produced.

This area of analysis was chosen as a person would control the device for a potentially extended period. If the device is front-heavy, or has a high weight in general, it will cause strain on the user, making the device harder to use. To minimize the load on a hand, the ideal device will be balanced over the center of mass of the handle, or as close to centered as possible. Alternatively, if required, a counterweight can be added instead of adjusting the location of the handle. However, this would increase the weight of the device, which is not desired.

Design Metrics

The applicable needs and specifications, as outlined in Table 1, all deal with the mass of the system and the offset of the overall center of mass (COM). The pitch moment about the wrist is the most important, as it deals with the safety of the user due to the potential for repetitive strain. The total print time of all of the 3D-printed components should be minimized to increase manufacturing efficiency.

Table 1. Needs and Specifications.

Index	Metric	Units	Target Value
N1	Net pitch moment about the wrist	N-m	< 0.15
N2	COM offset from grip axis	mm	< 15
N3	Total mass	kg	< 1.0
N4	Filament print time	hr	< 10

The analysis will aim to minimize each of these values, prioritizing N1 and N2. The mass of the system and the printed components can be adjusted as necessary, but will only be necessary if the system cannot be centered over the handle with ease.

Detailed Design

Figure 1 below shows the SolidWorks model for the handheld system, with the components in green being 3D-printed, and the remaining components being pre-existing technological components. Each has an indicated center of mass, and the red arrows show the freedom of location change for the handle, which will be determined based on the mass distribution analysis to minimize the load on the person holding the device.

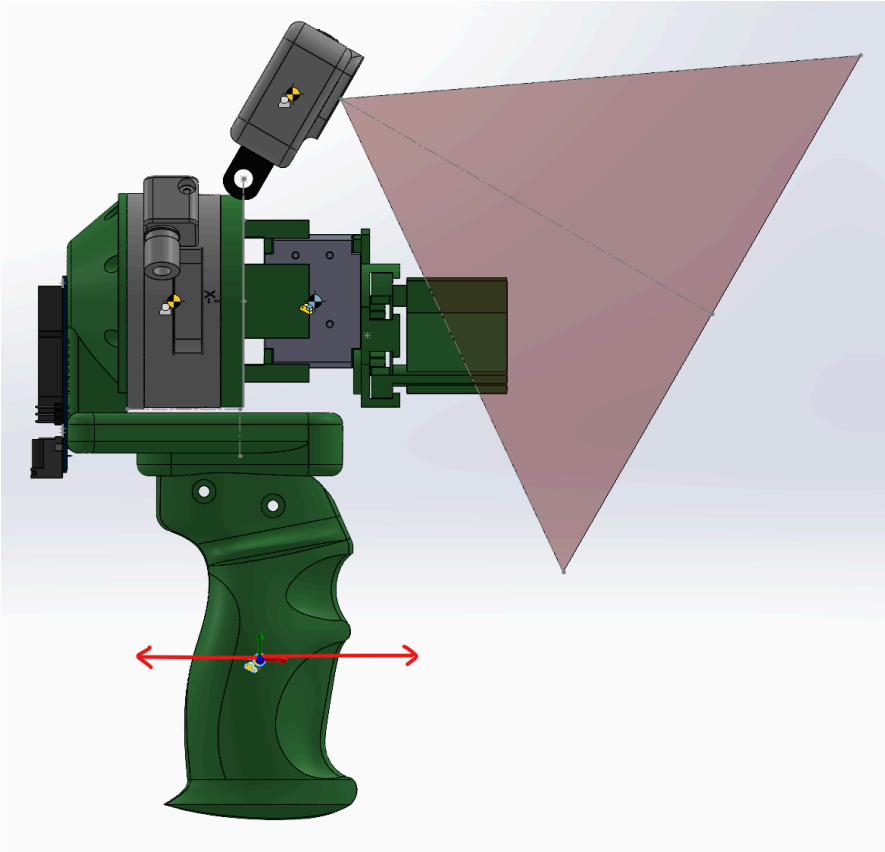


Figure 1. Model of handheld system.

From left to right, the gray components include the FT sensor, the GoPro, and the servo motor. The figure above also indicates the field of view of the GoPro with the red triangle. This property was used to determine a range of potential angles for the GoPro, with the front face being 60° from the vertical in the figure. The minimum angle for the GoPro was determined to be 45°.

Analysis and Evaluation

The weights and locations relative to the coordinate system based at the center of mass of the handle of each non-3D-printed component were first determined to complete the analysis. Table

2 contains each of these values, as well as the calculated moment that each applies. The GoPro was analyzed at its two extreme angles, being 45 and 60 degrees respectively. For this analysis, the weight of the 3D-printed components was considered negligible, as their masses are lower and the components are larger.

Table 2. Component parameters.

Component	**m (kg)**	**m-g (N)**	**x (mm)**	**m-g-x (N-m)**
FT Sensor	0.255	2.502	-30.01	-0.0751
GoPro @ 60°	0.126	1.236	+11.62	+0.0144
GoPro @ 45°	0.126	1.236	+18.70	+0.0231
Servo Motor	0.065	0.638	+19.32	+0.0123

The total weight of these components, regardless of the angle of the GoPro, is 4.375 N. Using the equation:

$$M_0 = \Sigma m_i \cdot g \cdot x_i$$

The total moment with the GoPro at 60° from the horizontal comes out to -0.0484 N-m. With the GoPro at 45°, this value is -0.0396 N-m.

Based on these values, it can be concluded that the FT sensor has the largest impact on the weight and moment of the system. It, on its own, overpowers the other two components on the opposing side of the handle.

When shifting the handle, each component's moment will be changed. To account for this, the equation below can be used.

$$M(\Delta) = \Sigma m_i \cdot g \cdot (x_i - \Delta)$$

$$M(\Delta) = \Sigma m_i \cdot g \cdot x_i - \Delta \cdot \Sigma m_i \cdot g$$

$$M(\Delta) = M_0 - W \cdot \Delta$$

Setting $M(\Delta) = 0$,

$$\Delta = \frac{M_0}{W}.$$

Δ indicates the new location of the handle's center of mass, relative to its initial location. With the GoPro at 60°, this shift is -11.1 mm. At 45°, the shift is -9.1 mm. Table 3 shows the comparison of the original moment to the adjusted moment.

Table 3. Comparison of moments.

GoPro angle	Moment before (N-m)	Moment after shift (N-m)	Reduction
60°	0.0484	0 (ideal)	100%
45°	0.0396	0 (ideal)	100%

Before the shift, the moment applied to the wrist can still be considered negligible. 0.0484 N-m is less than 1% of the maximum voluntary capacity, 8 N-m, not causing any noticeable fatigue. While a shift may not be necessary to keep the strain on a wrist negligible, it is still important to know that minor adjustments to the location of components or the weight of the 3D-printed components will not cause any major changes to the moment of the device.

In the alternative method of adding a counterweight, arbitrarily 60mm in front of the grip, the mass would need to be 0.082 kg (60° case) or 0.067 kg (45° case). Compared to the total weight of 0.466 kg, including the 3D-printed components, these values would incur a 15 to 18% increase in mass. Thus, it is less ideal to add the counterweight.

Broader Impacts

The primary broader impact of this design is user health and safety. A handheld device whose center of mass is offset from the grip axis imposes a continuous torque on the user's wrist and forearm as long as the device is held. For larger-scale data collection, the hold time increases. For the FT sensor, GoPro, and servo motor, the net moment about the original grip axis is 0.0484 N-M (GoPro at 60°) and 0.0396 N-m (GoPro at 45°), corresponding to center of mass offsets of 11.1 mm and 9.1 mm. Shifting the handle by these respective values eliminates the moment in the ideal case. Although the absolute moment is small compared to typical wrist strength, it acts continuously during sustained use, so removing most of it reduces cumulative load and the associated risk of fatigue and repetitive strain.

The design has also been determined to have a favorable manufacturing and economic profile. By balancing through shifting the handle, no additional mass is added, and thus no additional material must be used and purchased. In the case of adding a counterweight, a 15-18% increase of weight would be seen, also causing additional strain on the user unnecessarily.

The analysis is limited to the FT sensor, GoPro, and servo motor. The 3D-printed parts were excluded, and the analysis assumed the device is held level, so the vertical COM position and external loads are not captured.

References

https://pmc.ncbi.nlm.nih.gov/articles/PMC4676844/